

Comparison of Cylinder Count to Engine Displacement

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A client named Big Dawg has commissioned this work with the research question: Do cars with more cylinders not have bigger engines?

The goal of this report is to provide Big Dawg with clarity as to if cars with more cylinders have bigger or not.

This report will investigate the relationship between engine displacement and number of cylinders by using linear regression and the tools available on R studio.

Creating a Cylinder and Engine Displacement Table

```
# Calling the tidyverse package.  
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr    1.5.1  
## v ggplot2    3.5.2      v tibble     3.3.0  
## v lubridate  1.9.4      v tidyr      1.3.1  
## v purrr      1.1.0  
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# Calling mtcars and assigning it to the object mtcars.  
mtcars <- mtcars
```

```
# Creating Table 1 and assigning it to the object my_table. This involves selecting the columns cylinder and displacement.  
my_table <- mtcars |>  
  select(cyl, disp) |>  
  summarise(displacement = round(mean(disp), 2),  
            .by=cyl)
```

```

# Formatting my_table to match the lab instructions. Changing the names of the columns in Table 1 and u
my_table <- my_table |>
  rename(Cylinders = cyl,
         Displacement = disp_m)

knitr::kable(my_table, align = 'c', caption =
  "Mean engine displacement by number of cylinders") |>
  kableExtra::kable_styling(full_width = FALSE, position = "center")

```

Table 1: Mean engine displacement by number of cylinders

Cylinders	Displacement
6	183.31
4	105.14
8	353.10

Creating a Cylinder and Engine Displacement Bar Chart

```

# Creating and formatting the Bar Chart to align with the lab instructions.
my_table |>
  ggplot(aes(x = Cylinders, y = Displacement)) +
  geom_bar(stat = "identity") +
  labs(title = "Figure 1. Mean engine displacement by number of cylinders",
       x = "Number of Cylinders",
       y = "Engine Displacement (cm^3)") +
  theme_bw() +
  theme(
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.border = element_blank(),
    axis.line = element_line(color = "black")
  )
# Remove full panel border

```

Figure 1. Mean engine displacement by number of cylinders

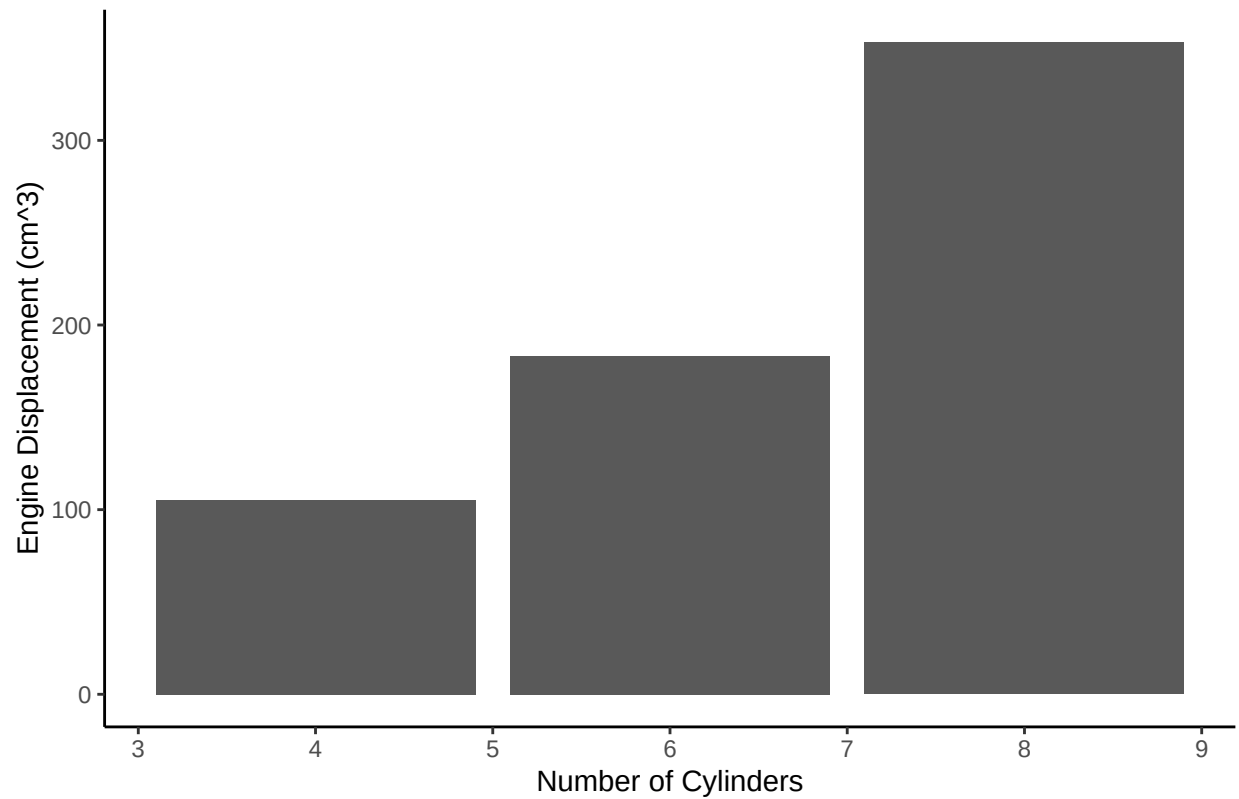


Figure 1 shows that as cylinders increase, engine displacement also increases. This shows that Big Dawgs original statement is false and cars with more cylinders do have bigger engines.